

Hitesh Kumar Gupta

Software Development Engineer · Graduating 2027
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EXPERIENCE

VaporFi Labs · Remote

Sep 2022 - Apr 2026

Software Engineer (Part-Time)

- Built event-driven indexers ingesting high-volume transaction streams into PostgreSQL in real time; designed composite indexes and tuned query plans to hold p95 read latency under 200 ms through sustained production load.
- Led end-to-end delivery of a marketplace handling 65,000+ assets, spanning smart contracts, distributed metadata storage and a React/Next.js frontend; raised Core Web Vitals from 63 to 95+ via an ISR/SSG refactor.
- Architected a multi-network routing aggregator, reverse-engineering undocumented closed-source logic to reconstruct an optimal path-selection algorithm minimising execution cost. Began as a broadly defined problem with no specification or documentation.
- Designed REST and GraphQL APIs with schema validation enforced at the service boundary; caching and request deduplication cut redundant network calls ~40%. Containerised services with Docker and ran CI/CD on self-hosted infrastructure.
- Audited smart contracts ahead of deploy; shipped asynchronously on a distributed remote team, every change going through peer code review.

Indian Institute of Technology Kharagpur · Dept. of Artificial Intelligence

May 2026 - Jun 2026

R&D Intern, under Dr. Koustav Rudra

- Cut a 100K-call distributed inference workload from 24 h to under 3 h (10x) on shared GPU infrastructure; profiled the job as prefill-bound, then redesigned memory allocation and batch scheduling to eliminate out-of-memory and overflow failures under contention.
- Built a fault-tolerant 8-stage data pipeline processing 1,100+ transcripts into 100K+ structured records; per-item incremental writes gave crash recovery and mid-job resumption across multi-hour compute runs with zero data loss.
- Designed a constrained partitioning scheme with an over-select-then-backfill solver meeting exact size targets under disjointness constraints; added embedding-based near-duplicate detection and an automated quality filter that cut the defect rate below 4%.
- Track Coordinator for the FIRE 2026 shared task (IIT Kharagpur x IISER Kolkata); the dataset ships as the track's official evaluation data.

Indian Institute of Technology (ISM) Dhanbad · Computer Science & Engineering

Feb 2026 - Apr 2026

R&D Project Intern, under Dr. Sachin Tripathi

- Implemented mutual authentication and key agreement across a three-tier distributed consortium network (cloud, fog nodes, edge devices) with ECC-based identity generation and a four-message handshake producing a verified shared session key.
- Built a fault-tolerant transaction system on a two-organisation consensus network: event-emitting application logic, a gateway listener ingesting external data and submitting signed transactions, and a role-separated web interface; demonstrated end-to-end state transition driven by external events. MeitY-funded (MeitY/2024-25/1106/CSE).

Birsa Institute of Technology, Sindri

Jun 2025 - Jul 2025

Deep Learning Intern, under Dr. Khustar Ansari

- Led 4 engineers translating research papers into deployable Python artifacts; defined shared standards for model-training pipelines and cloud deployment, reaching a 100% deployment success rate across all systems.

PUBLICATION

Gupta, H. K. (2025). *When Better Eyes Lead to Blindness: A Diagnostic Study of the Information Bottleneck in CNN-LSTM Image Captioning Models*. International Journal of Computer Applications. DOI: 10.5120/ijca2025925560. Cited 3 times, including in IEEE-published research, IOP Science and the NCIT Journal.

EDUCATION

K. K. College of Engineering & Management, Dhanbad

2027

B.Tech, Computer Science & Engineering · First Rank in programme

Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Databases, Distributed Systems, Computer Organisation, Discrete Mathematics, AI & Machine Learning, Information Retrieval, Data Mining.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), SQL, Bash, Solidity
CS Fundamentals: Data structures & algorithms, object-oriented design, operating systems, computer networks, complexity analysis
Distributed Systems: Event-driven architecture, fault tolerance, consensus protocols, message pipelines
Backend & APIs: Node.js, FastAPI, Express, REST, GraphQL, WebSockets
Databases: PostgreSQL, Redis, relational modelling, query optimisation & indexing, ETL pipelines
Cloud & Infrastructure: AWS, Docker, Kubernetes, CI/CD, Linux, Nginx, Git
ML: PyTorch, TensorFlow, Transformers, vLLM, Hugging Face

ACHIEVEMENTS

- First Rank in the CS & Engineering programme; Hindustan Talent Honour 2023.
- arXiv endorsement (cs.CV, cs.LG), obtained independently without institutional affiliation.
- Track Coordinator, cryptocurrency IR shared task, FIRE 2026 (IIT Kharagpur x IISER Kolkata).